

$$\boxed{2.43} \quad h = \frac{1}{n+1}, \quad x_k = k \cdot h$$

$$y_k = y(x_k)$$

$$y''(x_k) = \frac{y_{k+1} - 2y_k + y_{k-1}}{h^2}$$

$$y'' - y = 0$$

$$\Rightarrow \frac{y_{k+1} - 2y_k + y_{k-1}}{h^2} - y_k = 0 \quad \text{for } k=1, \dots, n$$

$$y_{k+1} - (2+h^2)y_k + y_{k-1} = 0 \quad \text{for } k=1, \dots, n$$

$$y_0 = 0 \Rightarrow -(2+h^2)y_1 + y_2 = 0$$

$$y_{n+1} = 1 \Rightarrow y_{n-1} - (2+h^2)y_n + 1 = 0$$

$$\Rightarrow A = \begin{bmatrix} -(2+h^2) & 1 & 0 & 0 \\ 1 & -(2+h^2) & 1 & 0 \\ 0 & 1 & -(2+h^2) & 1 \\ \vdots & & & \\ 0 & & & 1 & -(2+h^2) \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ -1 \end{bmatrix}$$

$$A \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = b$$